INDEX DEFINITION AND CALCULATIONS OF VALUE AND RETURNS

Learning Outcomes

- describe a security market index
- calculate and interpret the value, price return, and total return of an index

A **security market index** represents a given security market, market segment, or asset class. Most indexes are constructed as portfolios of marketable securities.

The value of an index is calculated on a regular basis using either the actual or estimated market prices of the individual securities, known as **constituent securities**, within the index. For each security market index, investors may encounter two versions of the same index (i.e., an index with identical constituent securities and weights): one version based on price return and one version based on total return. As the name suggests, a **price return index**, also known as a **price index**, reflects *only* the prices of the constituent securities within the index. A **total return index**, in contrast, reflects not only the prices of the constituent securities but also the reinvestment of all income received since inception.

At inception, the values of the price and total return versions of an index are equal. As time passes, however, the value of the total return index, which includes the reinvestment of all dividends and/or interest received, will exceed the value of the price return index by an increasing amount. A look at how the values of each version are calculated over multiple periods illustrates why.

The value of a price return index is calculated as:

$$V_{PRI} = \frac{\sum_{i=1}^{N} n_i P_i}{D} \qquad 1$$

where

V_{PRI} = the value of the price return index

n_i = the number of units of constituent security i held in the index portfolio

N = the number of constituent securities in the index

P_i = the unit price of constituent security i

D = the value of the divisor

The **divisor** is a number initially chosen at inception. It is frequently chosen so that the price index has a convenient initial value, such as 1,000. The index provider then adjusts the value of the divisor as necessary to avoid changes in the index value that are unrelated to changes in the prices of its constituent securities. For example, when changing index constituents, the index provider may adjust the divisor so that the value of the index with the new constituents equals the value of the index prior to the changes.

Index return calculations, like calculations of investment portfolio returns, may measure price return or total return. **Price return** measures only price appreciation or percentage change in price. **Total return** measures price appreciation plus interest, dividends, and other distributions.

Calculation of Single-Period Returns

$$PR_I = \frac{V_{PRI1} - V_{PRI0}}{V_{PRI0}} \quad (2)$$

where

PR_I = the price return of the index portfolio (as a decimal number, i.e., 12 percent is 0.12)

V_{PRI1} = the value of the price return index at the end of the period

V_{PRI0} = the value of the price return index at the beginning of the period

Similarly, the price return of each constituent security can be expressed as:

$$PR_i = \frac{P_{i1} - P_{i0}}{P_{i0}} \quad (3)$$

where

PR_i = the price return of constituent security *i* (as a decimal number)

P_{i1} = the price of constituent security *i* at the end of the period

P_{i0} = the price of constituent security *i* at the beginning of the period

Because the price return of the index equals the weighted average of price returns of the individual securities, we can write:

$$PR_I = \sum_{i=1}^{N} w_i PR_i = \sum_{i=1}^{N} w_i \left(\frac{P_{i1} - P_{i0}}{P_{i0}} \right) \quad (4)$$

where:

PR_I = the price return of index portfolio (as a decimal number)

PR_i = the price return of constituent security *i* (as a decimal number)

N = the number of individual securities in the index

w_i = the weight of security *i* (the fraction of the index portfolio allocated to security *i*)

P_{i1} = the price of constituent security *i* at the end of the period

P_{i0} = the price of constituent security *i* at the beginning of the period

Equation 4 can be rewritten simply as:

$$PR_I = w_1PR_1 + w_2PR_2 + \ldots + w_NPR_N \quad (5)$$

where

PR_I = the price return of index portfolio (as a decimal number)

PR_i = the price return of constituent security *i* (as a decimal number)

w_i = the weight of security *i* (the fraction of the index portfolio allocated to security *i*)

N = the number of securities in the index

Total return measures price appreciation plus interest, dividends, and other distributions. Thus, the **total return** of an index is the price appreciation, or change in the value of the price return index, plus income (dividends and/or interest) over the period, expressed as a percentage of the beginning value of the price return index. The total return of an index can be expressed as:

$$TR_I = \frac{V_{PRI1} - V_{PRI0} + Inc_I}{V_{PRI0}} \quad (6)$$

where

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V_{PRI0} = the value of the price return index at the beginning of the period

Inc_I = the total income (dividends and/or interest) from all securities in the index held over the period

The total return of an index can also be calculated as the weighted average of total returns of the constituent securities. The total return of each constituent security in the index is calculated as:

$$\mathrm{TR}_i = \frac{P_{1i} - P_{0i} + Inc_i}{P_{0i}} \quad (7)$$

where

TR_i = the total return of constituent security *i* (as a decimal number)

P_{1i} = the price of constituent security *i* at the end of the period

P_{0i} = the price of constituent security *i* at the beginning of the period

Inc_i = the total income (dividends and/or interest) from security *i* over the period

Because the total return of an index can be calculated as the weighted average of total returns of the constituent securities, we can express total return as:

$$\mathrm{TR}_I = \sum_{i=1}^{N} \mathrm{w}_i \mathrm{TR}_i = \sum_{i=1}^{N} \mathrm{w}_i \left(\frac{P_{1i} - P_{0i} + Inc_i}{P_{0i}} \right) \quad (8)$$

Equation 8 can be rewritten simply as

$$\mathrm{TR}_I = \mathrm{w}_1\mathrm{TR}_1 + \mathrm{w}_2\mathrm{TR}_2 + \ldots + \mathrm{w}_N\mathrm{TR}_N \quad (9)$$

where

TR_I = the total return of the index portfolio (as a decimal number)

TR_i = the total return of constituent security *i* (as a decimal number)

w_i = the weight of security *i* (the fraction of the index portfolio allocated to security *i*)

N = the number of securities in the index

Calculation of Index Values over Multiple Time Periods

The calculation of index values over multiple time periods requires geometrically linking the series of index returns. With a series of price returns for an index, we can calculate the value of the price return index with the following equation:

$$V_{PRIT} = V_{PRI0}(1 + \mathrm{PR}_{I1})(1 + \mathrm{PR}_{I2})\ldots(1 + \mathrm{PR}_{IT}) \quad (10)$$

where

V_{PRI0} = the value of the price return index at inception

V_{PRIT} = the value of the price return index at time *t*

PR_{IT} = the price return (as a decimal number) on the index over period *t*, *t* = 1, 2, …, *T*

For an index with an inception value set to 1,000 and price returns of 5 percent and 3 percent for Periods 1 and 2 respectively, the values of the price return index would be calculated as follows:

Period	Return (%)	Calculation	Ending Value
0		1,000(1.00)	1,000.00
1	5.00	1,000(1.05)	1,050.00
2	3.00	1,000(1.05)(1.03)	1,081.50

where

$V_{TRI\,0}$ = the value of the index at inception

V_{TRIT} = the value of the total return index at time t

TR_{IT} = the total return (as a decimal number) on the index over period t, t = 1, 2, …, T

Suppose that the same index yields an additional 1.5 percent return from income in Period 1 and an additional 2.0 percent return from income in Period 2, bringing the total returns for Periods 1 and 2, respectively, to 6.5 percent and 5 percent. The values of the total return index would be calculated as follows:

Period	Return (%)	Calculation	Ending Value
0		1,000(1.00)	1,000.00
1	6.50	1,000(1.065)	1,065.00
2	5.00	1,000(1.065)(1.05)	1,118.25

As illustrated above, as time passes, the value of the total return index, which includes the reinvestment of all dividends and/or interest received, exceeds the value of the price return index by an increasing amount.

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YG Kind of confused as per why in the practice question No 1, we use the sum of prices at the end of the period divided by the sum of the prices at the beggining minus one to calculate the return of the index...Should we not use the weight of each constituent security ?

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HX A friendly reminder: Do not do the practice problems, because many of them are solvable only when you learn next lesson: price index.

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